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Big arena, small potatoes: A mixed-methods investigation of atmospheric cues in live-streaming e-commerce

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ABSTRACT

Brands' self-built live streaming e-commerce has emerged as an embryonic and effective way of supporting customer purchase decisions. Moreover, the question of how small- and medium-sized enterprises can leverage IT-enabled factors to survive in the big arena has become a pressing concern. Extant research primarily focuses on broadcaster- and product-related factors by emphasizing psychological mechanisms and ignoring contextual factors from a holistic perspective. To address these research gaps, the complex interaction effects between IT-based atmospheric cues are explored in this paper using a mixed-methods approach. Drawing on affordance theory, we also integrate IT-based features with customer perceptions to further investigate the influencing mechanism of innovative decision-making contexts on customers' quick decisions.

1. Introduction

The dramatic increase in customer acquisition and conversion costs has triggered fierce competition in online markets, which has galvanized e-commerce firms to further develop and transform themselves [4]. Live streaming e-commerce (LSE) is an innovative type of e-commerce that features real-time visibility and synchronous interactivity characteristics. It has become an efficient channel for prompting customers' quick purchase decisions and for improving firms' economic benefits. Based on innovative live streaming technologies, LSE has transformed product presentations from textual and graphical descriptions to real-time interactive videos. This reduces the time required for customers to acquire and process information, thereby shortening their decision-making path. The global market size of LSE was valued at USD 50.11 billion in 2020 and is expected to expand at a compound annual growth rate (CAGR) of 21.0% from 2021 to 2028 [23], which constitutes an exceedingly impressive market prospect.

There are two major types of LSE [23]: (1) LSE hosted by key opinion leaders (KOLs) and (2) brand self-built LSE (BLSE), with a salesperson as the broadcaster. In KOL-based LSE, firms can entice a large number of customers to watch [42] as the social influence of KOLs means they can attract customer attention efficiently [20]. Further, they can generate celebrity and conformity effects [47], resulting in a massive number of impulsive purchases. However, most firms, especially small- and medium-sized enterprises (SMEs), cannot afford the extremely high cost of KOLs. Additionally, given that KOLs promote hundreds of brands within 2–3 h of live streaming, this provides insufficient brand communication for customers to create brand cognition and form deep brand experiences. For example, Austin Li is a Chinese KOL who advocates numerous brands and introduces each product in only 30 s [23]. In such a limited time, it is impossible for customers to systematically integrate all the information in the environment to make decisions. In this context, customer decisions are primarily based on trust/affection for KOLs and products [47]. Thus, although "people," "product," and "place" have always been the key retailing elements, existing studies on KOL-based LSE have only investigated the importance of people- and product-related factors in supporting customer decisions [47].

Compared with KOL-based LSE, firms eliminate middlemen (i.e., KOLs) in BLSE, focusing solely on promoting their own brand and significantly reducing their expenses. In this context, all environmental configurations (e.g., information sources and the category of promoting content) are directly controlled by the firms and are specifically designed for their own brands. Taking advantage of the synchronous interactivity and real-time visibility characteristics of live streaming, firms in the BLSE paradigm can directly offer authoritative shopping guidance and synchronously respond to customer comments in detail.

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This constitutes a bidirectional and direct information communication process between firms and customers. Moreover, customers can establish direct connections with firms by using and interacting with IT artifacts in BLSE. This also allows them to build relationships with the IT environment while dynamically accumulating brand experience. And the IT-enabled dynamic brand experience can also be regarded as customers' perceived use of BLSE, herein we refer to this as BLSE affordance. Importantly, diverging from KOL-based LSE, the social influence and reputation of broadcasters in BLSE are very limited. It is the IT-based "place/environment" of BLSE that constructs customer brand experience and becomes the major channel that connects customers with brands. This underscores the importance of contextual factors (atmospheric cues) in supporting customers' quick decisions. In addition, it is vital for small potatoes to leverage these cues to gain optimal improvements in economic benefits.

Existing research on BLSE is rather sparse. Moreover, previous studies on LSE have focused only on KOL-based LSE, investigating how factors related to people (i.e., KOLs and customers) [42] and products (e.g., product price and scarcity) affect the purchase intention and engagement behaviors of customers [20]. However, this ignores the impact of IT-based atmospheric cues in BLSE. With regard to the influencing mechanism, existing LSE studies are mainly based on the stimulus-organism-response (S-O-R) model, which recognizes information cues as stimuli that influence customers' psychological processing and subsequently influence their behavioral responses [15]. However, these investigations have only explored the impacts of LSE from psychological and emotional perspectives, ignoring IT-enabled experiences or the cognitive appraisal of customers. Importantly, BLSE is an IT-dependent decision aid context. Hence, it is necessary to combine IT features with customer perceptions to explain the complex decision-making process more effectively.

Motivated by these research gaps, this study aims to answer the following questions: (1) *What are the specific atmospheric cues in the BLSE context?* (2) *How can these context-specific factors influence customers' impulsive purchase decisions?* (3) *How can SMEs leverage the power of atmospheric cues to acquire competitive advantages and achieve commercial benefits?* Accordingly, we combine IT features with customer perceptions and experiences and draw on affordance theory to explore the complex decision-making processes in BLSE. Three potential contributions are expected from this research. First, we will conceptualize the atmospheric cues in BLSE, contributing to its understanding, connotation, and measurement in a new context. Second, the study will contribute to the decision support system (DSS) literature by investigating customer decision-making process in a real-time visualized and synchronous interactive environment, extending the understanding of the implementation of BLSE as a decision aid context. Third, this is one of the first studies to examine the BLSE context using a mixed-method approach, which can help achieve both scientific research realism and rigor through interpretive and positivist perspectives.

2. Literature review

2.1. Decision support system and live streaming e-commerce

DSS is one of the major areas in IS [5]. Existing research has primarily focused on the optimal design parameters and development processes for implementing DSS [2]. There are four essential sub-fields [16]: (1) personal decision support systems (e.g., financial planning systems); (2) group support systems (e.g., enterprise reporting and analysis systems); (3) intelligent decision support systems (e.g., natural language processing systems); and (4) knowledge management-based systems (e.g., knowledge-sharing community). Importantly, LSE is an innovative real-life decision context [15] that contains huge volumes of data/information, model (or decision aid) bases, and interfaces. These allow customers to gain product information and purchase recommendations in real-time to optimize their decisions. This paper is derived from previous DSS research, focusing on how firms can leverage IT-based atmospheric cues in BLSE to enhance customers' quick decisions (impulsive purchases), which contributes to extending the innovative implementation of DSS.

As mentioned previously, there are two major types of LSE: KOL-based LSE and BLSE. Although both of these IT environments are based on real-time visibility and synchronous interactivity technologies, they differ in terms of information sources (KOLs vs. employees of a brand) and the category of promoting content (multiple brands vs. a single brand), resulting in the differences in environmental configurations. Compared to KOL-based LSE, BLSE eliminates middlemen (i.e., KOLs), enabling firms to promote their own brand. In this case, firms can offer authoritative brand information directly, communicate with customers, and respond to comments and complaints quicker and more effectively. This improves the perceived reliability of information and the timeliness of feedback, enhancing direct connections between customers and firms (and the IT environment). In BLSE, instead of trusting broadcasters, customers rely more on integrating cues into the IT environment to generate brand experiences and make decisions. However, few LSE studies have differentiated between the two types of LSE, with the majority focusing on the context of KOL-based LSEs.

Existing LSE research has identified several antecedents to purchase intention [20,42], including factors related to KOLs (e.g., celebrity attractiveness and trustworthiness), customers (e.g., perceived values), and products (e.g., product credibility). However, such investigations have only emphasized people- and product-related factors in the KOL-based context, ignoring the significance of contextual elements (atmospheric cues) in BLSE. In particular, the question of how SMEs can allocate different environmental elements to support customers' quick decisions optimally remains starkly unexplored. With regard to influencing mechanisms, previous research has mainly investigated the impacts of LSE from psychological and emotional perspectives [47], employing the S-O-R model [15], engagement theory [42], and motivation/value theory [47]. However, BLSE is an IT-dependent context. Therefore, it is necessary to combine IT-based features with customer perceptions to further explain the complex decision-making process and to extend existing theoretical development.

2.2. Affordance theory

Affordance Theory combines the theories of perception and value. This provides a systematic interaction between the features of the IT environment and the actions to be taken in a specific context to explain how the environment is perceived and leads to actions [17]. Previous IS research has applied affordance theory to examine a wide array of information technologies [31], the effects of IT features on engagement behaviors [42], and design science [25]. Recently, this theory has been increasingly employed to investigate the relationship between the capabilities of shopping websites and their consequent actions [42]. For example, in the LSE context, Sun et al. [42] defined visibility, metavoicing, and guidance shopping as LSE affordances, implying that they can significantly affect customer purchasing intention. However, the definitions and dimensions of LSE affordances provided by the authors are still based on the research of Dong and Wang [11] in traditional social commerce, which neglects the real-time and no-middlemen-based characteristics of BLSE.

In line with extant research, we define BLSE affordance as customers' perceived properties of BLSE that could drive them to make decisions [31]. Unlike traditional e-commerce (which only provides customers with pictures, texts, and static experiences), BLSE integrates real-time presentations, co-viewing, and synchronous interaction functionalities. This allows customers to gain a more realistic and flowing experience in which to make decisions. There are also differences in response timeliness between the KOL-based LSE and BLSE. Within 2–3 h of live streaming, KOLs have to promote hundreds of brands with very limited introductions for each brand, with little time available for answering

customers' brand-related queries and concerns. In this situation, customers are unable to form a deep brand experience. By comparison, all environmental configurations are controlled by the firms themselves in BLSE, meaning that they have sufficient time to respond to comments, questions, and even complaints. This can enhance customers' brand cognition, enabling them to accumulate brand experiences dynamically. Thus, in this paper, we regard dynamic brand experience as a unique feature of BLSE that is distinct from previous research in the field.

2.3. Dynamic brand experience

Brand experience comprises customers' subjective internal responses (sensations, feelings, and cognitions) and behavioral responses (created by brand-related stimuli) [48]. During ongoing product presentations in BLSE, customers can continuously update their views on brands through real-time interactions. Similar to the dynamic nature of knowledge accumulation, brand experience is also an evolving process in this new context. According to Yu et al. [48], dynamism is an important attribute of brand experience and refers to "the changing cognitive states of interaction between various stakeholders at different stages of brand production and consumption". In BLSE, dynamic brand experience combines real-time interactive technologies with customers' perceptions/cognitions. Moreover, as the BLSE environment changes, customers' perceptions and cognitions of the products/brands will also change at different stages of product presentations. However, the empirical study of dynamic brand experience remains in its infancy, especially regarding the paradigms of the new variants of e-commerce. To provide a greater understanding of how this new affordance is perceived, we introduce the concept of atmospheric cues.

2.4. Atmospheric cues

Atmospheric cues are environmental elements that marketers can design to entice responses from customers [28]. This concept combines multiple environmental treatments and their interaction effects, emphasizing that customers process atmospheric characteristics holistically rather than piecemeal [26]. We introduce this concept into BLSE with the aim of investigating the influence of environmental stimuli from a more holistic perspective (i.e., interactive insights of environmental elements). Existing studies have investigated how customers behave when they encounter atmospheric cues (such as ambiance and information quality) in different shopping contexts (i.e., in-store and social commerce). Two types of atmospheric cues have been identified [43]: task and social. Task cues are elements that are utilitarian and relevant to task efficiency and effectiveness [43]. By comparison, social cues reflect the social presence of other customers and sales personnel in online marketplaces [43]. The specific dimensions and definitions of these atmospheric cues in any given environment vary depending on the context-specific attributes [26].

BLSE is a decision aid shopping context without any middlemen, where firms provide promotions, recommendations, and solutions directly to customers. Diverging from KOL-based LSE, all the contextual cues in BLSE are only associated with firms' own brand. Moreover, based on the innovative technologies, all these cues contain real-time visibility and synchronous interactivity features, which is also different from the atmospheric cues in traditional social commerce or in-store contexts. Importantly, as BLSE customers may have little connection and knowledge with the broadcasters or products of small potatoes, it is the virtual place/ environment that becomes the major channel connecting customers with the brand. Thus, it is vital for SEMs to be able to identify the specific contextual cues (atmospheric cues) in BLSE and further combine IT features with customer perceptions. This will allow exploration of how these cues can impact customers' dynamic brand experience and support their quick decisions to purchase.

To address these research gaps, we applied a mixed-method design to identify the specific atmospheric cues and the potential relationships among constructs. Fig. 1 presents how we embedded the constructs in the empirical model and the detailed procedures of the mixed-method design.

3. The mixed-methods design

To achieve scientific rigor and realism, a mixed-method design containing both qualitative and quantitative stages was applied in this paper [21,45,46]. This approach has several methodological advantages [45]: (1) it is greatly beneficial for producing a wider assortment of divergent and/or complementary views [46], which offers stronger inferences than a single method or worldview; (2) it is suitable for answering both confirmatory and explanatory research questions [46]; (3) it is suitable for exploring our new research context (i.e., the BLSE context), where issues are difficult to describe and explain using existing theories. In this research, a qualitative study (Study 1) was initially conducted to identify and discover special atmospheric cues in BLSE. These findings were then applied to develop hypotheses and a theoretical model in the quantitative study (Study 2). Accordingly, our theoretical model is based on Study 1 and existing theoretical underpinnings, which are assessed using construct development and validation and tested using the structural equation modeling technique. The specific procedures are presented in Fig. 1.

4. Study 1: Qualitative research

4.1. Data collection

The qualitative data were collected by the first and second authors from two sources: (1) two additional pilot interviews, which helped us to modify the interview guide, and (2) 35 semi-structured interviews with live-streaming customers. All interviews were conducted between April 2020 and June 2020, lasted for approximately 20 min each, and were transcribed within 24 h [13]. During these initial stages of the qualitative study, our questions focused on customer experiences when using BLSE and what they liked or did not like about BLSE concerning the atmosphere produced by SMEs. In the later stages, the focus of the questions shifted to emerging concepts and themes, paying particular attention to how customers discussed and evaluated brands in terms of the atmospheric conditions associated with live-streaming product broadcasting. To collect these interpretations, customers were encouraged to describe the characteristics of the atmospheric cues that influenced them most and were asked to elaborate on specific examples and feelings about the brands that promoted or reduced their impulsive purchasing intention.

4.2. Data analysis

The data analysis was an iterative process, adopting "the strategy between a prior and inductive" [33] and following the procedures suggested by Cheng et al. [7]. This strategy includes a "start list," which allows new related findings to be added to the list [7]. First, existing literature on the topics of atmospheric cues, live streaming, brand experiences, and affordance was reviewed and summarized to establish the conceptual foundation of atmospheric cues (i.e., guidance cues, bullet cues, and parasocial interactions). This constituted the start list in this study. The definitions of task and social cues provided by Tang and Zhang [43] were used to identify any specific atmospheric cues since these concepts have been used in customer behavior literature for several years [10,43]. Second, primary interview content related to our start list was further identified by repeatedly reading and comparing each transcript [7], as shown in Table 1. Third, to ensure the reliability and validity of the coding results, two authors (the first and the second author) coded the qualitative data, following Song et al. [41]. The first coded the interview data based on the list we concluded from the literature. Then, the second author, who is well trained and with

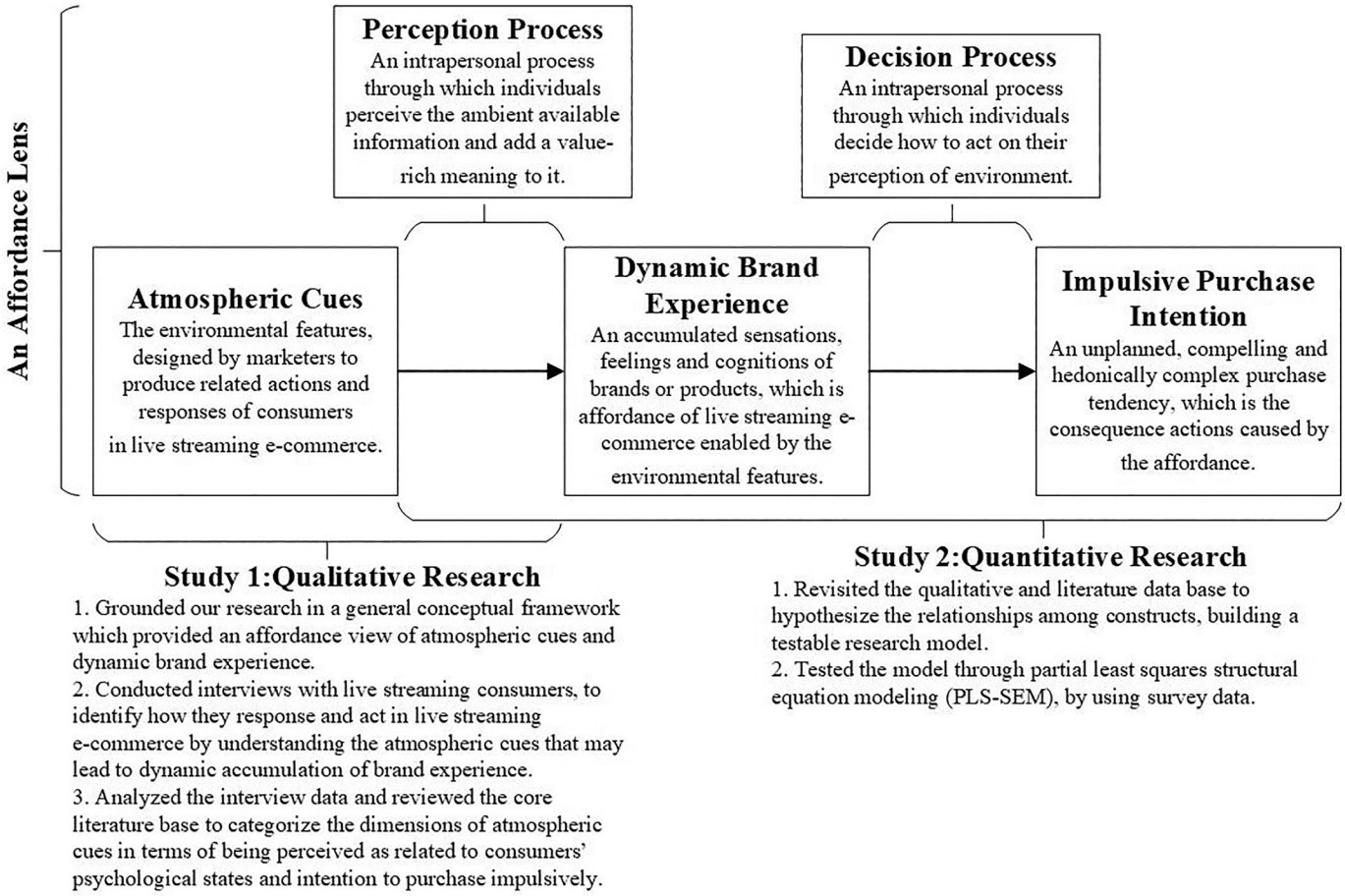


Fig. 1. The mixed-methods approach.

Table 1
Data supporting the theme "Atmospheric cues".

Associated Constructs	Representative Quotations
Guidance information	1."In BLSE, the task-related guidance (i.e., products functions or instructions) helps me a lot to make decisions." 2."… I mean the useful multi-perspective introduction of the product can help my decision."
Bullet information	1."In BLSE, the pop-up messages (i.e., 'who is going to buy XX') always make me feel like many people are rushing to purchase. This also impacts my purchase decisions." 2. "In BLSE, I can obtain other customers' opinions and questions from the bullet screen, which is supplementary information for my decision. It is very useful for me."
Parasocial interaction	1."The responses from broadcasters to customers' comments make me feel familiar with them, and I like the harmonious atmosphere very much." 2."Broadcasters' response reflects that he is listening to us and cares a lot about our requirements, which demonstrates their respect for us. Just like we were friends"

Table 2
Atmospheric cues for BLSE.

Atmospheric Cues	Times Mentioned	# Of People
Guidance information	33	26
Bullet information	25	23
Parasocial interaction	17	16

expertise in both coding and the research topic, repeated the same coding procedure independently. Between the two coders, agreement on the frequencies of each atmospheric cue was calculated (Table 2). We calculated the consistency of the two authors using Cohen's kappa embedded in NVivo 11. The final Cohen's kappa of 0.76 was well above the 0.70 thresholds, indicating acceptable inter-coder consistency between the two authors. After the 28 interviews were coded, no new codes emerged, which confirmed theoretical saturation [41].

The results revealed that there are two primary types and three dimensions of atmospheric cues in BLSE. These types were as follows: (1) task cues (guidance information) and (2) social cues (parasocial interactions and bullet information). Specifically, the cues were categorized as guidance information if they were provided by employees in real-time and appraised as being directly related to task accomplishment [43]. The cues were classified as bullet information if they appeared on the bullet screen and displayed other customers' brand-related questions/comments. Furthermore, cues were categorized as parasocial interactions if they were intimate, friend-like interpersonal connections that were unilaterally controlled by the employees [10].

We also found that task and social cues could either reinforce or weaken each other in the dynamic brand experience in BLSE. For example, some customers indicated that they paid more attention to multiple task cues (e.g., verbal and visual product information) and less attention to social cues (e.g., emotional interactions). However, others claimed that communication on the bullet screen enriched their consumption experience by providing supplementary shopping information. This indicates that there may be an interaction effect between the three atmospheric cues. Additionally, it is important to note that atmospheric cues refer to environmental stimuli, which are significantly associated with environmental components [28].

5. Building and hypothesizing the research model

5.1. Atmospheric cues and dynamic brand experience

5.1.1. Guidance information

In BLSE, guidance information is highly dependent on customers' personalized needs [42], allowing them to receive help from broadcasters immediately and directly. This can enhance interactions between customers and broadcasters [11], improving customers' perception of telepresence and social presence [42]. Moreover, it can improve customers' satisfaction and immersion in the environment [42], enhancing brand image to customers. Additionally, according to Mazaheri et al. [32], when customers are immersed in the context, their brand knowledge and mental simulations of the brand can also be significantly improved. Importantly, this can enhance customers' brand cognition, improving their brand experience [48]. Thus, in the context of BLSE, the following hypothesis was developed:

H1. Guidance information has a positive effect on customers' dynamic brand experience in BLSE.

5.1.2. Bullet information

In this study, bullet information refers to the content (e.g., communications and suggestions) generated by other customers on a bullet screen. This allows customers to gain the shopping experiences and opinions of others in real-time. Importantly, this can provide customers with numerous pieces of supplementary brand information that cannot be acquired from broadcasters [22], enhancing their indirect experience of the brand. Additionally, because bullet information is updated in real-time, it can also create a warm atmosphere with many people watching together. This situation can improve customers' affection toward the shopping environment [32], enhance a sense of immersion [42], and improve customers' brand memory and brand cognition [48]. Based on these arguments, we posit the following:

H2. Bullet information has a positive effect on customers' dynamic brand experience in BLSE.

5.1.3. Parasocial interaction

Parasocial interaction is a one-sided, friend-like communication atmosphere generated by broadcasters and perceived by customers. It can build an intimate atmosphere/connection between customers and broadcasters [44], improving customers' perceptions of the credibility and trustworthiness of broadcasters [6]. This can significantly enhance customers' perceived value of the brand [22], improve the affection and cognition of the brand, and enhance customers' psychological immersion in the environment [6]. Accordingly, customers' cognitive appraisals of a brand can also be improved [48]. Therefore, the following hypothesis is presented:

H3. Parasocial interactions have a positive effect on customers' dynamic brand experience in BLSE.

5.2. Two-way interaction effect of atmospheric cues

BLSE is a complex IT-based shopping environment, where all atmospheric stimuli (guidance information, bullet information, and parasocial interactions) are simultaneously integrated into one scene. According to the biased-competition model of attention, when there are two or more stimuli in a visual display, different types of stimuli may have interaction effects on each other, as individuals' cognitive capacity is limited [24]. In the following section, the interaction effects between task cues (guidance information) and two different types of social cues (parasocial interactions and bullet information) are discussed.

5.2.1. The role of guidance information in parasocial interactions

According to cue utilization theory [29], in shopping environments that contain multiple cues, customer judgments and choices are based on the combination of these different types of cues. Although parasocial interactions represent a type of social cue that can enhance shopping experiences [10], when considering other cues, previous conclusions could change. The extent to which a specific cue is utilized in assessing products varies with its diagnosticity [36], meaning the perceived reliability and relativity of cues in meeting individuals' requirements. In BLSE, the major motivation of customers is to accomplish shopping tasks [23], as demonstrated by Customer_04: "*I don't deny social factors will also impact my decisions, but it is still the product information that matters for me and my evaluations on the brands…*". It should be noted that guidance information is closely associated with customers' shopping tasks and requirements. Moreover, compared to parasocial interactions, it is more diagnostic for customers when they understand and evaluate a brand. The more guidance information the IT environment contains the greater the likelihood that customers will expend more effort on processing these task cues and less effort on parasocial interaction. Thus, we posit the following:

H4a. In BLSE, guidance information negatively moderates the relationship between parasocial interactions and dynamic brand experience; when guidance information is high, the effect of parasocial interactions on dynamic brand experience is weakened.

5.2.2. The role of guidance information in bullet information

Unlike one-sided parasocial interactions, bullet information represents a two-side/bilateral synchronized communication process, which is another social cue in this study. Moreover, the interaction effect of guidance information on bullet information may also differ from the effect on parasocial interaction. Customer_06 described the following: *"When watching broadcasters' product presentations, I notice that there exist other customers' questions and complaints. And these cues are vital for me to make decisions."* According to the cue diagnosticity framework [36], compared to neutral or positive information, negative complaints and queries are more diagnostic for customers. In BLSE, guidance information is provided by employees, which is generally neutral. In this case, bullet information (which contains negative complaints and questions from other customers) could be more diagnostic. Additionally, although multiple task information (e.g., broadcasters' verbal recommendations and visual presentations) in BLSE can enhance the mental simulation, it also increases the workload on processing for customers. Accordingly, because customers' cognitive capacity is limited, the more guidance information the environment contains, the greater the likelihood that a customer cognitive overload will occur, rendering bullet information more diagnostic. Based on these arguments and evidence, we posit the following:

H4b. In BLSE, guidance information positively moderates the relationship between bullet information and dynamic brand experience; when guidance information is high, the effect of bullet information on dynamic brand experience is strengthened.

5.3. Three-way interaction effect of atmospheric cues

According to parasocial interaction theory, a customer's sense of immersion can be enhanced when they feel intimate with broadcasters [10]. Customer_07 stated that *"When I meet the broadcasters I like, I am more likely to immerse in the communication atmosphere, and enjoy their product presentations"*. This sense of immersion can improve customers' perceptions of environmental stimuli [42], enhancing brand identification and perceived brand value [20]. This can improve customers' understanding of the information delivered in the BLSE environment. Additionally, when customers are immersed, their degree of pleasure can be significantly improved [42]. This can enhance customers' acceptance of both bullet and guidance information, consolidating the effects of these cues. Based on these arguments, we posit the following:

H5. There is a three-way interaction effect between guidance information, bullet information, and parasocial interactions. Here, the positive moderating effect of guidance information on the relationship between bullet information and dynamic brand experience is stronger when parasocial interactions are higher and more sensitive to information and changes in the environment.

5.4. Dynamic brand experience and impulsive purchase intention

Numerous brand experience studies have confirmed that brand experience can enhance customers' perceived value of products, improve brand trust [12], and stimulate customers' impulsive purchase intention [25]. In BLSE, dynamic brand experience manifests as the summation of the accumulated experience of a brand. This can enhance customers' impressions of the brand, improve perceived brand credibility, and stimulate impulsive purchase intention [48]. Based on these arguments, we posit the following:

H6. In BLSE, dynamic brand experience has a positive effect on a customer's impulsive purchase intention.

Fig. 2 presents the research model and hypothesized relationships.

6. Study 2: Quantitative research

6.1. Measures

Existing scales were revised to measure the five reflective constructs and fit our research context. There were five reflective constructs: guidance information, measured by 4 items from Kim and Park [27]; bullet information, measured by 4 items from Shi and Liao [39]; parasocial interaction, measured by 8 items from Thorson and Rodgers [44]; dynamic brand experience, measured by 25 items from Yu et al. [48]; and impulsive purchase, measured by 4 items from Beatty and Ferrell [3]. All the items in this study were measured on a five-point Likert scale ranging from 1 (*strongly disagree*) to 5 (*strongly agree*).

6.2. Data collection

Our data were collected through an online survey distributed to customers who had engaged in BLSE in China in 2020. The link to the survey was distributed via the Wenjuanxing website. To control for social desirability bias (SDB), all data were collected through an anonymous self-reporting approach with socially acceptable questions. Thus, the outcome was based on internalized values that were independent of social considerations.

To calculate a suitable sample size, a power analysis was conducted using G*power 3.1 software [14]. The results suggested that a sample size of 74 would be sufficient for a one-tailed test and a power (1 – β) of 0.95, given α = 0.05. We received 575 responses, of which 118 were removed, as they reported no experience of using BLSE. A total of 457 usable responses (46.6% male and 53.4% female) were included in the empirical analysis. This gender distribution is consistent with extant research on BLSE [42,47]. The demographic distribution covered ages ranging from >16 to <18 (n = 11, 2.41%) to >45 (n = 16, 3.51%), with the majority of the respondents being in the age range 19–25 years (n = 234, 51.32%). Their average shopping frequency per month ranged from <3 times (n = 289, 63.38%) to >7 times (n = 34, 7.46%). Moreover, 136 respondents shopped in BLSE 4–7 times per month, accounting for 29.82% of the overall data. The average monthly income of respondents ranged from <2000 RMB (n = 115, 25.22%) to >8000 RMB (n = 52, 11.40%), while the majority of the respondents earned 4500–8000 RMB (n = 160, 35.09%).

6.3. Data analysis and results

6.3.1. Measurement model

We selected partial least squares structural equation modeling (PLS-SEM), which makes no assumptions about data distributions and allowed us to test the quality of the measurement models and the research hypotheses [18]. Based on an iterative method, PLS-SEM maximizes the explained variance of endogenous constructs [19]. In addition, this method can handle problematic modeling issues in the social sciences, such as complex models and unusual data characteristics [19].

Initially, a confirmatory factor analysis (CFA) was conducted. The Cronbach's alpha and composite reliability (CR) for each construct were > 0.7, and their average variance extracted (AVE) was >0.5. These results indicate reliable and convergent validity [18,19]. In addition, we tested discriminant validity (Table 3) through the Fornell–Larcker criterion, cross-loading, and the Heterotrait–Monotrait ratio of correlations. The mean squared variance (MSV) values were all less than the AVE, and the HTMT values all ranged from 0.321 to 0.737, indicating good discriminant validity [4,19].

Table 3
Discriminant validity: Fornell-Larcker criterion and HTMT ratio of correlations.

Construct	IPT	DBE	PSI	BI	GI
IPT	**0.875**	*0.523*	*0.357*	*0.321*	*0.330*
DBE	0.484	**0.720**	*0.737*	*0.592*	*0.623*
PSI	0.313	0.660	**0.829**	*0.408*	*0.519*
BI	0.286	0.531	0.351	**0.827**	*0.583*
GI	0.296	0.563	0.451	0.504	**0.887**

Note: Diagonal elements (bold) are the square root of AVE for each construct. Below-diagonal elements are the criterion of the Fornell-Larker, above-diagonal elements are the criterion of the HTMT (in italics).

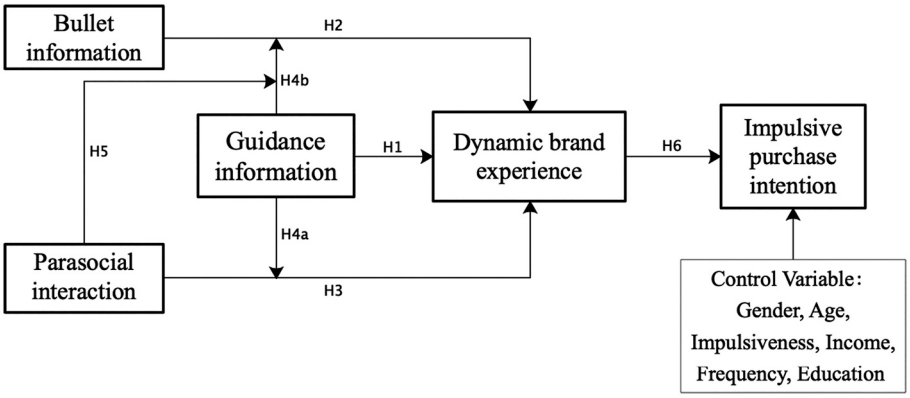


Fig. 2. Research model.

6.3.2. Common method bias

To control common method bias (CMB), we adopted two approaches: (1) optimizing the design of the study's procedures [21,35] and (2) conducting statistical controls [35,37]. First, when designing the procedures for Study 2, we attempted to minimize CMB by reducing evaluation apprehension and protecting respondent anonymity. Second, we conducted a PLS marker variable approach, following the procedures suggested by Rönkkö and Ylitalo [37]. First, we identified the number of live-streaming platforms used as a marker variable and then added this variable to the research model as an exogenous variable to predict each endogenous construct. Our results indicated that there were no significant correlations between the marker variable and other constructs in this study. Moreover, after adding the marker variable, the significant paths of the baseline model remained significant, indicating that our results were not affected by CMB [37].

6.3.3. Hypotheses testing

To test H1–H6, we conducted two correlation tests—one for the baseline test and one for the three-way interaction effect of atmospheric cues and dynamic brand experience (DBE). First, we examined the mediating effect of DBE between atmospheric cues (guidance information, GI; parasocial interaction, PSI; and bullet information, BI) and impulsive purchase intention (IPT) through SEM-PLS. We then conducted a multiple regression analysis to test the two- and three-way interaction effects [30].

Fig. 3 shows the test results of our baseline model, as well as the controlled variables, indicating that the atmospheric cues in BLSE were positively related to DBE. Specifically, PSI ($\beta = 0.465$; Cohen's f2 = 0.382), BI ($\beta = 0.249$; Cohen's f2 = 0.105), and GI ($\beta = 0.226$; Cohen's f2 = 0.079) on DBE were statistically significant ($p < .001$), indicating support for H1–H3. Meanwhile, the DBE ($\beta = 0.298$; Cohen's f2 = 0.140) also had significant ($p < .001$) and positive effects on IPT, supporting H6. The results of the multiple regression analysis are presented in Table 4. This indicated that the two-way interaction effects on which we focused were significant, with GI × BI ($\beta = 0.106$, $p < .001$), and GI × PSI ($\beta = -0.178$, $p < .001$) on DBE, supporting H4a and H4b. Additionally, the three-way interaction was significant, with GI × BI × PSI ($\beta = 0.071$, $p < .05$) on DBE, and the incremental variance (explained by the three-way interaction effect) was 3.2%, supporting the prediction in H5. As summarized by the results in Table 5, all hypotheses were supported.

6.3.4. Post hoc analysis

To examine the significance of the interaction effects between atmospheric cues and DBE in greater detail, we plotted these interaction effects by entering unstandardized regression coefficients (including the intercept) and means and standard deviations for the independent variables [1], as shown in Figs. 4–7. Fig. 4 presents that high GI (illustrated by the solid line) has greater effects on BI-DBE than the low GI (dotted line), indicating that GI positively moderated the relationship between BI and DBE. However, as shown in Fig. 5, the slope was steeper when GI was low (dotted line), suggesting that GI negatively moderated the relationship between PSI and DBE [9]. Figs. 6–7 display the three-way interaction effect of GI, BI, and PSI on DBE. As shown in Fig. 6, with high PSI, BI was positively related to DBE, regardless of whether GI was high or low. However, as shown in Fig. 7, with low PSI, the positive relationship between BI and DBE was stronger, and the slope in situations of high GI (solid line) was sharper. This implies that the reinforcement effect of GI on the relationship between BI and DBE was enhanced when PSI was high.

We also conducted a mediation analysis following the procedures suggested in previous studies [40]. As presented in Table 6, we obtained

Table 4
The results of the regression analyses.

Variable	Model 0	Model 1	Model 2	Model 3
Control variables	**β**	**β**	**β**	**β**
Age	−0.008	−0.013	−0.017	−0.012
Education	0.136***	0.001	0.012	−0.001
Gender	0.196***	0.013	−0.004	−0.021
Income	−0.095	−0.047	−0.027	−0.053
Impulsiveness	0.480***	0.220***	0.167***	0.187***
Frequency	−0.002	0.001	0.546	0.014
Main effects				
Guidance information		0.224***	0.230***	0.138***
Bullet information		0.262***	0.312***	0.294***
Parasocial interaction		0.462***	0.365***	0.383***
Two-way interaction				
Guidance information × Bullet information			0.106***	0.137*
Parasocial interaction × Bullet information			0.003	0.050
Guidance information × Parasocial interaction			−0.178***	−0.256***
Three-way interaction				
Guidance information × Bullet information × Parasocial interaction				0.071*
R^2	0.084	0.589	0.623	0.650
F	7.313***	34.404***	9.169***	23.451***
ΔR^2	0.084	0.505	0.034	0.027
Mean VIF	2.298	2.219	2.125	2.097

Notes: *p < .05, **p < .01, ***p < .001.

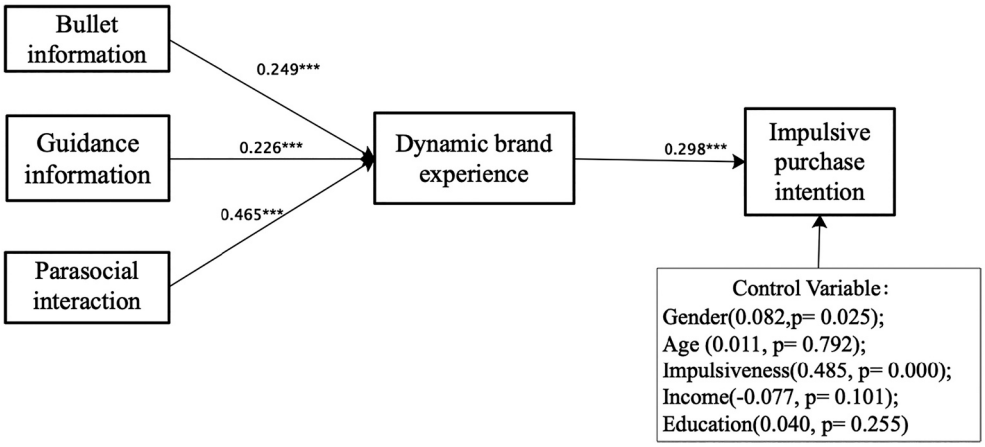


Fig. 3. Results of the baseline model.

Table 5
The overall hypotheses testing results.

Hypothesis	Path	Path coefficients	Standard Deviation	T-values	*P*-Values	Supported?
H1	GI → DBE	0.226	0.053	4.254	0.000	Yes
H2	BI→DBE	0.249	0.051	4.885	0.000	Yes
H3	PSI → DBE	0.465	0.048	9.764	0.000	Yes
H4a	BI x GI → DBE	0.087	0.040	2.151	0.032	Yes
H4b	PSI x GI → DBE	−0.256	0.055	4.670	0.000	Yes
H5	BI x PSI x GI → DBE	0.078	0.033	2.368	0.018	Yes
H6	DBE → IPT	0.298	0.053	5.595	0.000	Yes

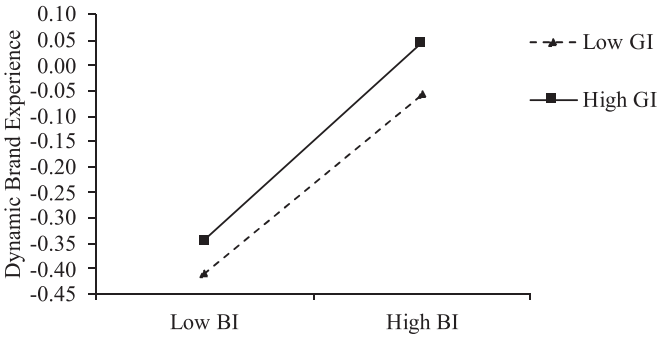


Fig. 4. Moderating effects of GI on BI-DBE.

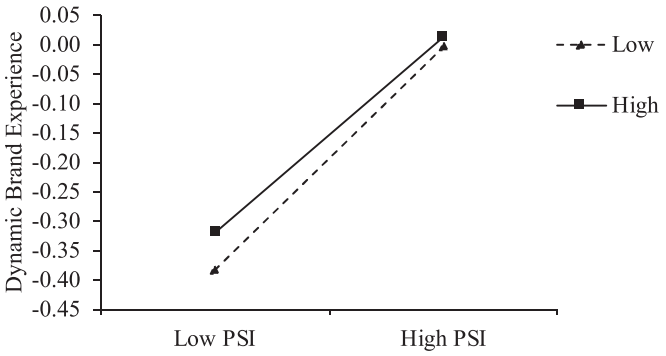


Fig. 5. Moderating effects of GI on PSI-DBE.

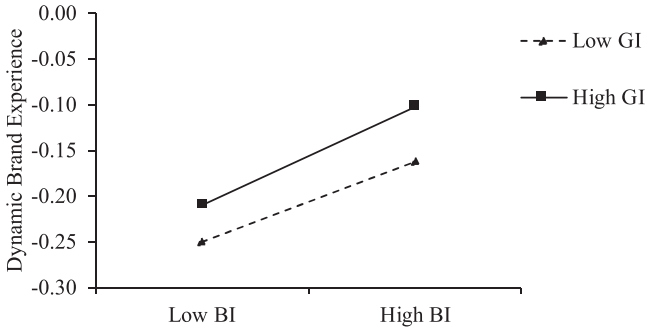


Fig. 6. The three-way interaction effect among atmospheric cues at a high level of PSI.

5000 resamples and conducted 95% bootstrap percentile intervals for both indirect and direct effects and the results of the mediation testing. The effects of guidance information, bullet information, and parasocial interaction were all partially mediated by DBE.

6.3.5. Robustness checks

To ensure stability and validity, we conducted robustness checks following procedures from Dwivedi et al. [12] and Sarstedt et al. [38]. The FIMIX-PLS approach was selected to test the robustness of our theoretical model [8,34]. First, after specifying the effects of additional demographic influences (i.e., age, education, and income) on impulsive purchase intention, the results mimic those reported in Table 5, supporting the robustness of the structural model results [12]. Next, the results of FIMIX-PLS (Table 7) did not unambiguously point to a specific segmentation solution for two reasons: (1) the AIC3 and CAIC pointed to different segment numbers [19], and (2) the AIC_4 and Bayesian information criteria (BIC) suggested different segment numbers [38]. Thus, we conclude that our theoretical model is robust [38].

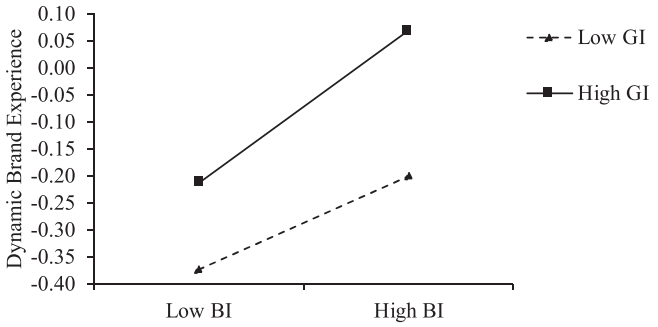


Fig. 7. The three-way interaction effect among atmospheric cues at a low level of PSI.

6.4. Summary of findings

The findings from both the qualitative and quantitative studies answered our research questions. With regard to the first question (What are the specific atmospheric cues in BLSE environments), both the task cue (guidance information) and social cues (bullet information and parasocial interactions) were identified through the iterative analysis of Study 1. For the second question (How can the cues influence customers' impulsive purchase intention), our results from Study 2 indicate that atmospheric cues can significantly affect customers' impulsive purchases by enhancing their probability of dynamically accumulating brand experience. For the third question, the findings of Study 1 and 2 indicate that the impacts of atmospheric cues on customers are interactive rather than piecemeal. Given the findings from analyzing interaction effects, to maximize brand experience accumulation, SMEs should pay more attention to maintaining the illusory intimacy relationship between broadcasters and customers. Furthermore, they should reinforce the acceptance of a high-level guidance information environment and enhance the influence of bullet information on improving customer experiences.

7. Implications and conclusions

7.1. Theoretical implications

First, our study contributes to the LSE literature by examining the

Table 6
Mediating effect of dynamic brand experience.

Variable		Mediation Test (ab)			Full/Partial Median Test (c')			Type of mediation
IV	MV	2.5% lower bound	97.5% upper bound	zero included?	2.5% lower bound	97.5% upper bound	zero included?	
PSI	DBE	0.378	0.533	No	0.041	0.118	No	Partial
GI		0.113	0.323	No	0.032	0.109	No	Partial
BI		0.156	0.359	No	0.095	0.192	No	Partial

Table 7
Fit indices for the one- to five-segment solutions.

	Number of segments				
Criteria	1	2	3	4	5
AIC (Akaike's Information Criterion)	1870.04	1824.53	1793.51	1698.68	**1661.09**
AIC_3 (Modified AIC with Factor 3)	1886.04	1857.53	1843.51	1765.68	**1745.09**
AIC_4 (Modified AIC with Factor 4)	1902.04	1890.53	1893.51	1832.68	**1829.09**
BIC (Bayesian Information Criteria)	**1936.04**	1960.65	1999.75	1975.04	2007.56
CAIC (Consistent AIC)	**1952.04**	1993.65	2049.75	2042.04	2091.56
HQ (Hannan Quinn Criterion)	1896.04	1878.15	1874.74	1807.53	**1797.56**
MDL_5 (Minimum Description Length with Factor 5)	**2328.02**	2769.11	3224.68	3616.45	4065.45
LnL (LogLikelihood)	−919.021	−879.267	−846.756	−782.341	**−746.543**
EN (Entropy Statistic (Normed))	na	0.271	0.611	0.674	**0.708**
NFI (Non-Fuzzy Index)	na	0.309	0.603	0.612	**0.630**
NEC (Normalized Entropy Criterion)	na	332.938	177.903	149.092	**133.658**

complex interaction effects between different types of atmospheric cues. This provides a more general and holistic perspective for investigating how firms can optimally leverage IT-based contextual cues to enhance customers' quick decisions. BLSE is an integrated context that combines customer comments, purchase recommendations, and personalized services simultaneously into one scene. Previous research has only examined the single-factor influences of KOL- and product-related factors in KOL-based LSE [42]. However, this ignores the fact that in BLSE, the IT environment is the major channel connecting customers with the brand. Furthermore, IT-based contextual elements interactively affect customers' decisions rather than piecemeal. Our results confirm the two- and three-way interaction effects between the atmospheric cues in BLSE, underscoring the necessity to integrate parasocial interactions, guidance information, and bullet information simultaneously to support customers' quick decisions optimally. This goes beyond existing studies by offering deeper and fresher insights into understanding customers' decision-making processes in a new context.

Second, this paper contributes to LSE research by introducing affordance theory and the biased competition model of attention to the BLSE context. Existing research has primarily investigated the impacts of atmospheric cues from a psychological perspective, with the majority drawing on the S-O-R model. However, BLSE is highly dependent on synchronous communication technologies. Accordingly, considering only customers' emotions and perceptions may not fully explain their complex decision-making processes in BLSE. This study combines the feelings of customers pertaining to IT-related features (synchronous communication and real-time visibility) while regarding DBE as a special affordance of BLSE. This offers new insights into the influencing mechanisms of atmospheric cues in a new context.

Finally, this paper extends the existing DSS research by investigating complex customer decision-making processes in a real-life decision context (BLSE). Previous studies have primarily focused on optimal design parameters and development processes for implementing DSS [2]. However, unlike other online shopping contexts, BLSE is an decision aid context with real-time interactivity, synchronous interactivity, and no-middlemen-based features. Our results identified three context-specific atmospheric cues in BLSE (parasocial interaction, guidance information, and bullet information), confirming that these factors can interactively affect customer decisions. This expands the understanding of atmospheric cues in BLSE and contributes to extending the innovative implementation of DSS.

7.2. Managerial implications

This study revealed several practical suggestions for SMEs in operating BLSE. First, our results indicate that in BLSE, where guidance information is at a high level, the positive influence of customers' parasocial interactions on their accumulation of brand experience is reduced. This implies that the guidance information provided by employees is more diagnostic for customers in BLSE. In such cases, SMEs should focus more on improving their own employees' competencies and techniques when presenting products, delivering brand-related knowledge, and solving customers' shopping questions. For example, this could include conducting systematic education and training for employees.

Second, our results also revealed that with high guidance information, the positive impact of bullet information on DBE was strengthened in BLSE. As a result, SMEs need to place greater emphasis on encouraging customers to use the bullet screen to submit questions, comments, and even complaints. For instance, to improve customers' perceptions of brand service and brand memory, every item on a bullet screen should be responded to sequentially in real-time, paying special attention to negative information and complaints. Further, encouraging customers' to communicate on bullet screens contributes to the establishment of a positive brand image for other participants in the same BLSE.

Finally, our findings revealed that the effect of guidance information increased in strength when the broadcasters in SMEs were more likely to be considered friends by customers. This means that cultivating friendships between SME employees and their customers is more effective in enhancing the shopping experience than hiring celebrities as salespeople. Accordingly, SMEs should focus more on enhancing broadcasters' ability to mobilize customer emotions and on increasing perceived reciprocity and trustworthiness, increasing customers' sense of immersion, and maximizing their attitudes toward and experiences of the brand.

7.3. Limitations and future research

This study also has several limitations. First, all our constructs were from a self-reported approach, which is less accurate than using actual purchase data. Future research should pay more attention to real transactional data to study customer behavior more accurately. Second, this study was conducted in China, where national and cultural settings

are different from Western countries. Future studies are expected to expand our research model by adding cultural elements as a boundary condition to explore whether this model works in other cultural contexts. Additionally, this research identified three dimensions of atmospheric cues in BLSE, while the influence of other underlying cues (i.e., multisensory cues and page guidance) will become increasingly prominent as the popularity of virtual reality and 5G technology increases. Hence, future research should further expand the dimensions of atmospheric cues in BLSE. Finally, this paper assumes that the bullet information from other customers can be more objective and useful in helping users assess a product. However, different customers have various shopping experiences, which can result in subjective cognition bias in their product evaluations. Therefore, whether (and how) cognition bias from other customers can affect users' decisions in BLSE requires further exploration in future research.

Credit author statement

All authors equally contributed to the design and development of this research.

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